CASE STUDY ON

HOW SENSOR-BASED TECHNOLOGIES AND TRACKING SYSTEMS CAN BE USED TO AUTOMATE HIGHLIGHT GENERATION IN CRICKET MATCHES.

Problem Statement:

Manual highlight creation has several limitations:

- It requires significant time and human effort.
- Editors may miss important moments due to fatigue or time constraints.
- Real-time highlight generation is difficult.
- Large volumes of video data must be reviewed.

Therefore, an automated system is needed to analyze match data and identify key moments efficiently.

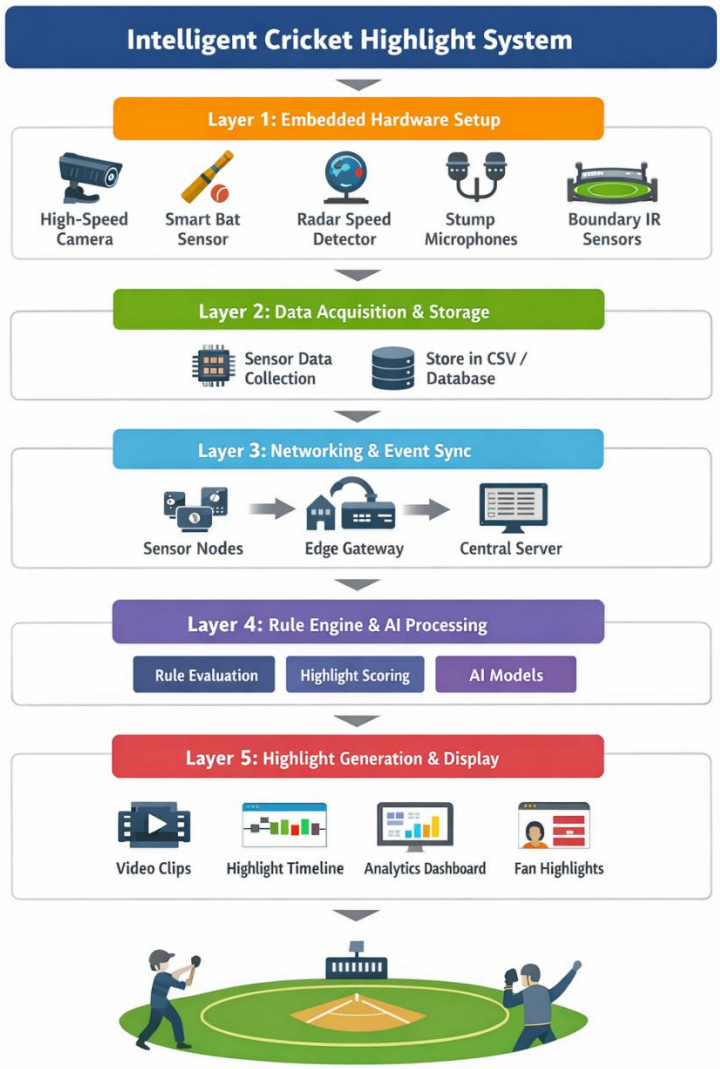
Proposed Solution:

- **Smart Sensor Integration:** Deploy high-speed cameras, smart bat sensors, radar speed detectors, stump microphones, and boundary IR sensors to capture real-time match events.
- **Automated Data Capture:** Continuously collect and convert sensor signals into structured digital data with timestamps.
- **Edge-Based Networking:** Use sensor nodes, an edge gateway, and a central server to synchronize and transmit data in real time.
- **Rule + AI Hybrid Engine:** Combine rule-based logic (boundary, wicket, six detection) with AI models to identify and rank highlight events.
- **Highlight Scoring Mechanism:** Assign an excitement score to events based on factors like runs, speed, player action, and crowd impact.
- **Automatic Video Extraction:** Link detected events with video timestamps to generate highlight clips instantly.
- **Interactive Analytics Dashboard:** Display match insights, event timelines, and performance analytics for analysts and broadcasters.
- **Fan-Ready Content Delivery**: Generate short highlight clips optimized for broadcast, streaming platforms, and social media sharing.

5 LAYERED APPROACH

Layer 1: Embedded Hardware Setup

- Captures real-time match events
- Detects ball speed, bat contact, and boundaries
- Acts as the data sensing layer of the stadium



Layer 2: Data Acquisition & Storage

- Collects data from all sensors
- Converts signals into structured digital data
- Stores match events for further processing

Layer 3: Networking & Event Synchronization

- Connects sensors to the central system
- Synchronizes all match events in real time
- Ensures accurate timing of every delivery

Layer 4: Rule Engine & AI Processing

- Analyzes collected sensor data
- Detects wickets, boundaries, and key moments
- Acts as the decision-making brain

Layer 5: Highlight Generation & Display

- Converts detected events into highlight clips
- Generates timelines and analytics dashboards
- Delivers engaging highlights for fans and broadcasters

LAYER 1: EMBEDDED DEVICES LAYER

Fielding Effort Tracking Sensors

Sensor Type	Device Used	Placement	Parameter Measured	Units
UWB Position Tag	DWM1001 Module	Player Vest	Player Position (X, Y)	meters
IMU Accelerometer	MPU6050 IMU	Player Vest	Acceleration	m/s^2
GPS Module	NEO-6M GPS	Player Vest	Latitude, Longitude	degrees
Derived Metric	Calculated	Software	Player Speed	m/s
Derived Metric	Calculated	Software	Distance Covered	meters

Ball Tracking Sensors

Sensor Type	Device Used	Placement	Parameter Measured	Interface with / Processor
Radar Speed Gun	Stalker Pro II	Behind Bowler	Ball Speed	km/h
High-Speed Camera	Phanton VEO	Stadium Roof	Ball Trajectory	coords
Optical Camera	4K Broadcast Cam	Field Angles	Ball Path	pixels
Pitch Camera	Overhead Camera	Above Pitch	Pitch Location	meters

Edge / Contact Detection Sensors

Sensor Type	Device Used	Placement	Parameter Measured
UltraEdge Microphone	MAX4466 Mic	Stump Area	dB
Thermal Camera	FLIR Lepton	Pitch Side	°C

Thermal Camera		Placement / Base	
FLIR Lepton	FLIR Lepton		dB
Thenmemeter	Raspberry Pi		°C

Boundary Detection Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
IR Sensor	IR Beam Sensor	Boundary Line		EDP
Optical Camera	HD Camera	Near Boundary	meters	EDP

Impact / Wicket Detection Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Stump Sensor	Vibration Sensor	In the Stumps	Voltage	ESP32
Bail Sensor	Accelerometer	Inside Bails	Voltage	ESP32
Piezo Sensor	Piezo Disc	Stump Base	Voltage	ESP32

Bat Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Bat IMU	MPU6050 IMU	Bat Handle	m/s^2	ESP32
Piezo Sensor	Piezo Sensor	Bat Blade	Newton	ESP32
Ultrasonic Sensor	HC-SR04	Near Bat	cm	ESP32
GPS Module	NEO-GM GPS	Bat Handle	coords	ESP32

Environmental Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Wind Sensor	Anemometer	Roof Top	m/s	ESP32
Temperature Sensor	DHT22	Side Wall	°C	ESP32
Decibel Meter	DHT22	Side Wall	°C	ESP32

Crowd Sound Detection Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Stump Sensor	Vibration Sensor	In the Stumps	Voltage	ESP32
Bail Sensor	Accelerometer	Inside Bails	kom	ESP32
Piezo Sensor	Piezo Disc	Stump Base	Notuxa	ESP32

Crowd Sound Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Microphone	MAX466 Mic	Stand Poles	dB	ESP32
Sound Sensor	KY-038 Module	Stands	Bmary	ESP32
Decibel Meter	Sound Level Meter	Media Box	dB	Raspberry Pi

Vibration Sensors

Sensor Type	Device Used	Placement	Units	Interface with / Processor
Piezo Vibration	Piezo Sensor	Growmt / Stumps	Voltage	ESP32
Accelerometer	MPU6050	Stump Area	m/s'	ESP32

1. Fielding Effort Sensors
- Used to track player position, speed, and movement patterns during fielding.

2. Ball Tracking Sensors
- Used to measure ball speed, trajectory, and pitch location throughout the delivery.

3. Edge / Contact Detection Sensors
- Used to detect bat–ball contact events through sound and thermal variations.

4. Boundary Detection Sensors
- Used to identify whether the ball has crossed the boundary line.

5. Impact / Wicket Detection Sensors
- Used to detect stump hits, bail movement, and wicket impact events.

6. Bat Sensors
- Used to measure bat swing dynamics such as acceleration, rotation, and impact force.

7. Crowd Sound Sensors
- Used to capture audience noise levels and excitement during gameplay.

8. Environmental Sensors
- Used to monitor external conditions like wind speed and temperature.

9. Vibration Sensors
- Used to measure vibrations caused by ball impacts on equipment or pitch.

LAYER 2: DATA ACQUISITION

Key Functions

• Collects real-time data generated by different sensors placed on players, equipment, and around the stadium.
• Converts raw sensor signals into structured numerical parameters.
• Organizes the data for each ball event into a dataset row.
• Stores the data in CSV or Excel format for easy analysis and rule engine processing.
• Maintains a chronological record of match events for further analytics.

Role in the System

The dataset generated in this layer becomes the primary input for the rule engine, which analyzes these parameters to determine whether a particular ball delivery should be considered a highlight moment.

DATASHEET USED: CRICKETDATA

Parameters Stored from Sensors

Parameter	Description	Sensor Source
ball_number	Unique identifier for each ball	Match event counter
over	Over number in the match	Match data
ball_speed	Speed of the bowled ball	Doppler radar
bat_acceleration	Bat acceleration during swing	MPU6050 IMU
bat_rotation_speed	Bat rotational velocity	Gyroscope sensor
impact_force	Force generated during bat-ball impact	Piezoelectric sensor
shot_speed	Speed of the ball after the shot	Radar / camera
shot_angle	Direction or elevation of the shot	High-speed camera
edge_detected	Whether a bat edge occurred	UltraEdge microphone
edge_sound_level	Sound intensity of the edge	Microphone sensor
boundary_crossed	Whether the ball crossed the boundary	IR boundary sensors
shot_distance	Distance travelled by the ball	Optical tracking cameras
fielder_speed	Speed of the nearest fielder	UWB + IMU sensors
distance_covered	Distance run by fielder	Player tracking sensors
reaction_time	Time taken for fielder reaction	Motion sensors
crowd_noise_level	Crowd noise intensity	Microphone sensors
crowd_spike	Sudden crowd excitement spike	Sound detection module
wind_speed	Wind speed affecting ball movement	Wind sensor
temperature	Ambient temperature	Temperature sensor
stump_hit	Whether the ball hit the stumps	Stump impact sensor
stump_vibration	Vibration intensity in the stumps	Vibration sensor

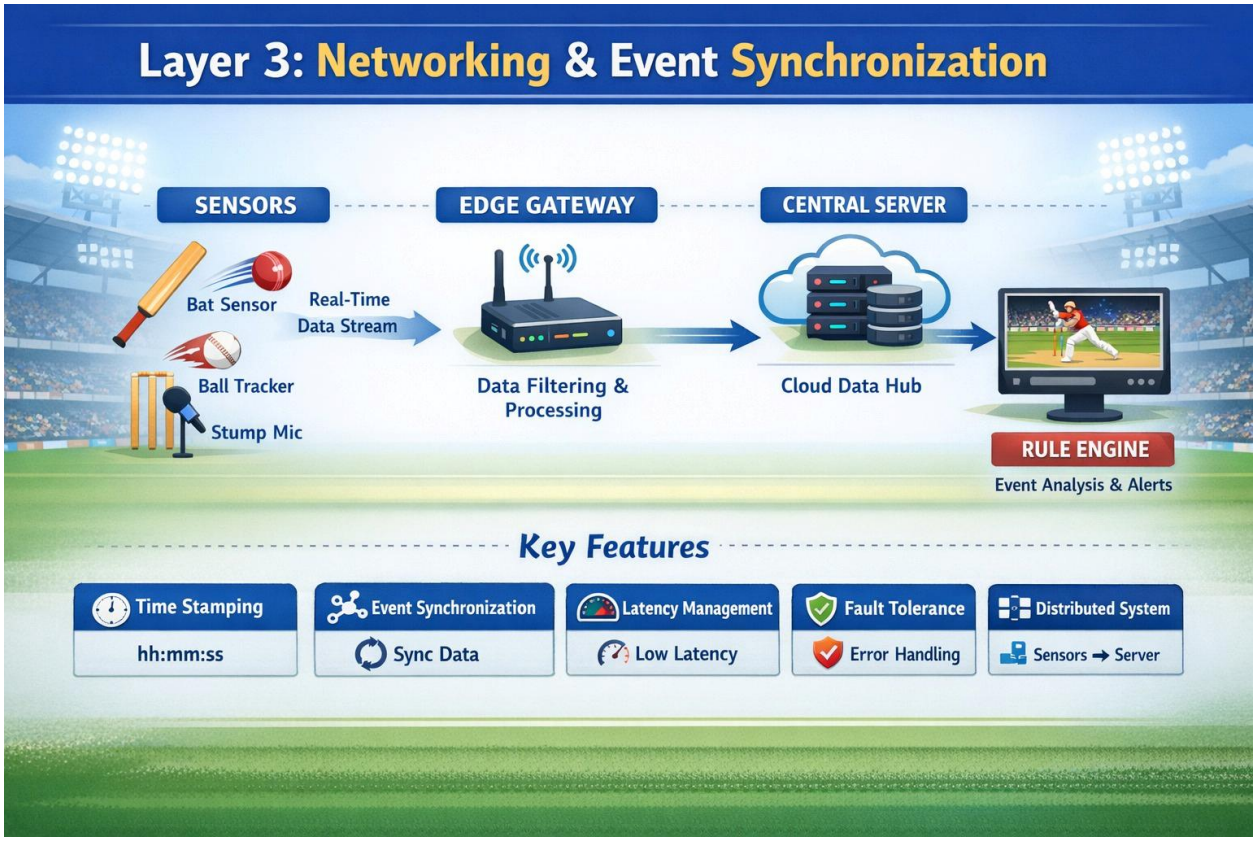
LAYER 3: NETWORKING AND EVENT SYNCHRONIZATION LAYER

Key Functions

- Transfers real-time data from sensors and gateways to the central server.
- Uses communication protocols like MQTT and TCP/IP for efficient data transmission.
- Synchronizes data from multiple sensors into a single ball event.
- Assigns timestamps to maintain correct sequence of events.
- Ensures continuous and reliable data flow during the match.

Role in the System

This layer acts as the communication backbone of the system, ensuring that all sensor data is accurately transmitted and synchronized in real time. It prepares well-aligned and time-ordered data for the rule engine, enabling correct analysis and highlight detection without delays or mismatches.



LAYER 4: RULE ENGINE

Rule Engine is the decision-making core that automatically identifies important moments in a cricket match using sensor data.

ALL ABOUT OUR RULE ENGINE:

→ 3 APPROACHES FOR 3 DIFFFERENT NEEDS

1. Overall Match Highlights

- Summarizes entire match
- Includes best moments (wickets, sixes, key turns)
- Generated after match ends

2. Mid-Match Highlights

- Shows important events during live match
- Updates after each over or session
- Keeps viewers engaged in real-time

3. Highlighted Moments in Match

- Focuses on single key events
- Example: one six, wicket, or stunning shot
- Instant detection and display

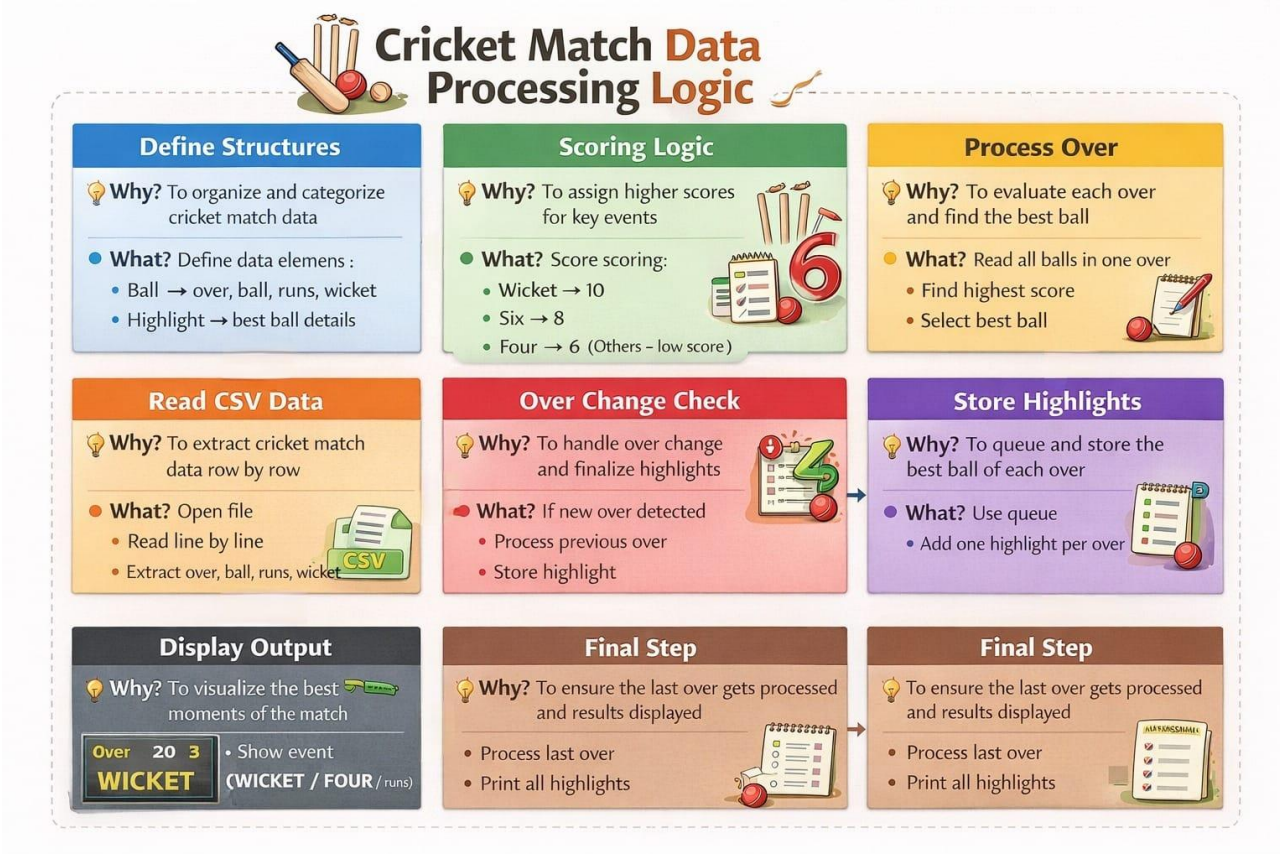
1.OVERALL MATCH HIGHLIGHTS

This approach analyzes each ball using multiple parameters such as shot quality, physics, fielding effort, and audience reaction and instead of simple rules (like only runs/wickets), it uses a multi-factor scoring model to detect high-impact events.

OVERALL IDEA:

- Read match data from CSV
- Analyse each ball using sensor-like parameters
- Assign a highlight score
- Classify events based on importance
- Select only premium highlights

CODE LINK: [OVERALLHIGHLIGHTSCODE]



SIMULATED OUTPUT:

```
C:\WINDOWS\system32\cmd.
Over 1 Ball 1 | Score: 250 | Highlight: STANDARD
Over 1 Ball 2 | Score: 260 | Highlight: STANDARD
Over 1 Ball 3 | Score: 270 | Highlight: STANDARD
Over 1 Ball 4 | Score: 170 | Highlight: MINOR
Over 1 Ball 5 | Score: 310 | Highlight: PREMIUM
Over 1 Ball 6 | Score: 220 | Highlight: STANDARD
Over 2 Ball 7 | Score: 480 | Highlight: PREMIUM
Over 2 Ball 8 | Score: 260 | Highlight: STANDARD
Over 2 Ball 9 | Score: 360 | Highlight: PREMIUM
Over 2 Ball 10 | Score: 370 | Highlight: PREMIUM
Over 2 Ball 11 | Score: 350 | Highlight: PREMIUM
Over 2 Ball 12 | Score: 350 | Highlight: PREMIUM
Over 3 Ball 13 | Score: 220 | Highlight: STANDARD
Over 3 Ball 14 | Score: 220 | Highlight: STANDARD
Over 3 Ball 15 | Score: 310 | Highlight: PREMIUM
Over 3 Ball 16 | Score: 260 | Highlight: STANDARD
Over 3 Ball 17 | Score: 270 | Highlight: STANDARD
Over 3 Ball 18 | Score: 290 | Highlight: STANDARD
Over 4 Ball 19 | Score: 220 | Highlight: STANDARD
Over 4 Ball 20 | Score: 220 | Highlight: STANDARD
Over 4 Ball 21 | Score: 220 | Highlight: STANDARD
Over 4 Ball 22 | Score: 340 | Highlight: PREMIUM
Over 4 Ball 23 | Score: 350 | Highlight: PREMIUM
Over 4 Ball 24 | Score: 270 | Highlight: STANDARD
Over 5 Ball 25 | Score: 220 | Highlight: STANDARD
Over 5 Ball 26 | Score: 290 | Highlight: STANDARD
Over 5 Ball 27 | Score: 220 | Highlight: STANDARD
Over 5 Ball 28 | Score: 220 | Highlight: STANDARD
Over 5 Ball 29 | Score: 340 | Highlight: PREMIUM
Over 5 Ball 30 | Score: 350 | Highlight: PREMIUM
Over 6 Ball 31 | Score: 220 | Highlight: STANDARD
Over 6 Ball 32 | Score: 250 | Highlight: STANDARD
Over 6 Ball 33 | Score: 380 | Highlight: PREMIUM
Over 6 Ball 34 | Score: 220 | Highlight: STANDARD
Over 6 Ball 35 | Score: 220 | Highlight: STANDARD
Over 6 Ball 36 | Score: 220 | Highlight: STANDARD
Over 7 Ball 37 | Score: 190 | Highlight: STANDARD
Over 7 Ball 38 | Score: 330 | Highlight: PREMIUM
Over 7 Ball 39 | Score: 290 | Highlight: STANDARD
Over 7 Ball 40 | Score: 270 | Highlight: STANDARD
Over 7 Ball 41 | Score: 310 | Highlight: PREMIUM
```

```
C:\WINDOWS\system32\cmd.
Over 1 Ball 5 | Score: 310 | PREMIUM HIGHLIGHT
Over 2 Ball 7 | Score: 480 | PREMIUM HIGHLIGHT
Over 2 Ball 9 | Score: 360 | PREMIUM HIGHLIGHT
Over 2 Ball 10 | Score: 370 | PREMIUM HIGHLIGHT
Over 2 Ball 11 | Score: 350 | PREMIUM HIGHLIGHT
Over 2 Ball 12 | Score: 350 | PREMIUM HIGHLIGHT
Over 3 Ball 15 | Score: 310 | PREMIUM HIGHLIGHT
Over 4 Ball 22 | Score: 340 | PREMIUM HIGHLIGHT
Over 4 Ball 23 | Score: 350 | PREMIUM HIGHLIGHT
Over 5 Ball 29 | Score: 340 | PREMIUM HIGHLIGHT
Over 5 Ball 30 | Score: 350 | PREMIUM HIGHLIGHT
Over 6 Ball 33 | Score: 380 | PREMIUM HIGHLIGHT
Over 7 Ball 38 | Score: 330 | PREMIUM HIGHLIGHT
Over 7 Ball 41 | Score: 310 | PREMIUM HIGHLIGHT
Over 8 Ball 48 | Score: 310 | PREMIUM HIGHLIGHT
Over 9 Ball 51 | Score: 350 | PREMIUM HIGHLIGHT
Over 10 Ball 56 | Score: 340 | PREMIUM HIGHLIGHT
Over 10 Ball 57 | Score: 310 | PREMIUM HIGHLIGHT
Over 10 Ball 59 | Score: 390 | PREMIUM HIGHLIGHT
Over 11 Ball 64 | Score: 310 | PREMIUM HIGHLIGHT
Over 15 Ball 88 | Score: 360 | PREMIUM HIGHLIGHT
Over 16 Ball 92 | Score: 310 | PREMIUM HIGHLIGHT
Over 17 Ball 97 | Score: 330 | PREMIUM HIGHLIGHT
Over 18 Ball 105 | Score: 340 | PREMIUM HIGHLIGHT
Over 20 Ball 117 | Score: 310 | PREMIUM HIGHLIGHT
Over 23 Ball 135 | Score: 340 | PREMIUM HIGHLIGHT
Over 24 Ball 139 | Score: 310 | PREMIUM HIGHLIGHT
Over 24 Ball 143 | Score: 310 | PREMIUM HIGHLIGHT
Over 26 Ball 153 | Score: 320 | PREMIUM HIGHLIGHT
Over 26 Ball 155 | Score: 340 | PREMIUM HIGHLIGHT
Over 27 Ball 161 | Score: 330 | PREMIUM HIGHLIGHT
Over 29 Ball 174 | Score: 310 | PREMIUM HIGHLIGHT
Over 31 Ball 184 | Score: 380 | PREMIUM HIGHLIGHT
Over 31 Ball 185 | Score: 330 | PREMIUM HIGHLIGHT
Over 32 Ball 192 | Score: 360 | PREMIUM HIGHLIGHT
Over 37 Ball 219 | Score: 310 | PREMIUM HIGHLIGHT
Over 38 Ball 223 | Score: 310 | PREMIUM HIGHLIGHT
Over 39 Ball 229 | Score: 310 | PREMIUM HIGHLIGHT
Over 39 Ball 231 | Score: 330 | PREMIUM HIGHLIGHT
Over 39 Ball 232 | Score: 310 | PREMIUM HIGHLIGHT
Over 40 Ball 235 | Score: 350 | PREMIUM HIGHLIGHT
```

2.MID MATCH HIGHLIGHT

Mid-match highlights provide real-time updates of important moments as the game progresses. They capture key events like wickets, boundaries, and turning points after each over, helping viewers stay engaged without waiting for the match to finish.

Overall Idea

- Read match data from CSV
- Process **each over separately**
- Pick **best ball (highlight)** per over
- Store and display highlights

CODE LOGIC:

1.Define Structures

- Ball → stores data of each ball
  (over, ball no, runs, wicket)
- Highlight → stores selected best moment

2. **Scoring Function (getScore)**
Assigns importance to each ball:

- Wicket → 10 (highest)
- Six → 8
- Four → 6
- Others → lower score

  ➞ Helps decide **best moment**

3. **Process One Over (processOver)**

- Input: all balls of one over
- Loop through each ball
- Calculate score using getScore()
- Pick ball with **maximum score**

  ➞ Return as highlight

4.Reading CSV File

- Open match.csv
- Skip header
- Read line by line
- Extraxct: over,ball,runs,wicket

5.Detect Over Change

- If current over ≠ previous over:
  - Process previous over
  - Store highlight in queue
  - Display live highlights
  - Clear data for next over

6. Store Highlights

- Use **queue**
- Each over → one highlight added

7. Display Function

- Prints all highlights in order
- Shows: Over & ball number &
  Event (WICKET / SIX / FOUR / runs)

8.**Final Step**

- Process last over
- Display complete highlight timeline

CODE LINK:[MIDMATCHAPPROACHCODE]

SIMULATED OUTPUTS:

```
=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)

=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)

=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)

=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)
Over 5 → Ball 26 (WICKET)

=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)
Over 5 → Ball 26 (WICKET)
Over 6 → Ball 35 (WICKET)
```

```
=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)
Over 5 → Ball 26 (WICKET)
Over 6 → Ball 35 (WICKET)
Over 7 → Ball 39 (SIX)
Over 8 ⬜⬜ Ball 43 (WICKET)
Over 9 → Ball 54 (SIX)
Over 10 → Ball 55 (SIX)
Over 11 → Ball 65 (WICKET)
Over 12 → Ball 68 (SIX)
Over 13 → Ball 75 (WICKET)

=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)
Over 5 → Ball 26 (WICKET)
Over 6 → Ball 35 (WICKET)
Over 7 → Ball 39 (SIX)
Over 8 ⬜⬜ Ball 43 (WICKET)
Over 9 → Ball 54 (SIX)
Over 10 → Ball 55 (SIX)
Over 11 → Ball 65 (WICKET)
Over 12 → Ball 68 (SIX)
Over 13 → Ball 75 (WICKET)
Over 14 → Ball 82 (WICKET)
```

```
=== LIVE HIGHLIGHTS ===
Over 1 → Ball 1 (FOUR)
Over 2 → Ball 11 (SIX)
Over 3 → Ball 15 (SIX)
Over 4 → Ball 21 (SIX)
Over 5 → Ball 26 (WICKET)
Over 6 → Ball 35 (WICKET)
Over 7 → Ball 39 (SIX)
Over 8 → Ball 43 (WICKET)
Over 9 → Ball 54 (SIX)
Over 10 → Ball 55 (SIX)
Over 11 → Ball 65 (WICKET)
Over 12 → Ball 68 (SIX)
Over 13 → Ball 75 (WICKET)
Over 14 → Ball 82 (WICKET)
Over 15 → Ball 86 (SIX)
Over 16 → Ball 93 (WICKET)
Over 17 → Ball 99 (WICKET)
Over 18 → Ball 104 (FOUR)
Over 19 → Ball 111 (FOUR)
Over 20 → Ball 117 (WICKET)
Over 21 → Ball 121 (FOUR)
Over 22 → Ball 131 (WICKET)
Over 23 → Ball 136 (FOUR)
Over 24 → Ball 144 (EDGE)
Over 25 → Ball 147 (FOUR)
Over 26 → Ball 151 (SIX)
Over 27 → Ball 160 (SIX)
Over 28 → Ball 168 (SIX)
Over 29 → Ball 169 (SIX)
```

3.HIGHLIGHT MOMENTS

Top highlight clips aim to capture the most exciting moments of the match, selecting events that stand out due to extraordinary performance, milestones, or dramatic occurrences. Each clip represents a single ball or event that has maximum impact, producing a concise highlight reel for viewers.

OVERALL IDEA:

- Read match data from a CSV file (ball-by-ball).
- Assign a highlight score to each ball based on rules for excitement and milestones.
- Sort balls by highlight score to identify the most important moments.
- Select top clips as individual highlights.
- Display clips sequentially, showing over, ball number, and score of that ball.

CODE LOGIC:

1. Define Structures

- BallEvent: Stores over, ball number, runs, wicket, boundary, shot distance, ball speed, shot speed, impact force, crowd noise, score, duration.
- Player: Tracks cumulative runs for milestone detection (half-century / century).

2. Scoring Function

- Assigns importance to each ball using rules:
    - Wicket → highest score
    - Century milestone → very high score
    - Half-century milestone → high score
    - Six → high score
    - Four → medium score
    - Fast shot/ball, hard hit, crowd excitement → medium score
- Total score = sum of all triggered rules.

3. Process Balls

- Loop through all balls from CSV.
- Compute highlight score for each ball.
- Track cumulative runs per player.

4. Read CSV File

- Open match.csv, skip header, read each line.
- Extract: over, ball, runs, wicket, boundary, speeds, impact, crowd.

5. Select Top Clips

- Sort balls by highlight score descending.
- Pick top balls as individual highlight clips.

6. Store Highlights

- Store each selected clip in a list/queue.
- Each clip represents one exciting moment.

7. Display Highlights

- Print sequentially: Over, Ball, Score / Event (WICKET / SIX / FOUR / milestone).
- Clips are standalone, easy to follow.

8. Final Step

- Process all remaining balls.
- Display complete highlight timeline with separate clips.

CODE LINK:[SELECTIVE EVENTS]

SIMULATED OUTPUT:

```
===== TOP 5 HIGHLIGHT CLIPS =====

--- Clip 1 ---
Over 31 Ball 185 | Score: 300.00

--- Clip 2 ---
Over 4 Ball 23 | Score: 270.00

--- Clip 3 ---
Over 2 Ball 12 | Score: 270.00

--- Clip 4 ---
Over 5 Ball 30 | Score: 270.00

--- Clip 5 ---
Over 31 Ball 184 | Score: 270.00

C:\Users\vinot\Downloads>
```

LAYER 5: DISPLAY OR OUTPUT LAYER

Key Functions

- Presents processed match highlights in a clear and user-friendly format
- Maintains a chronological timeline of important events (over-wise display)
- Enhances visibility by tagging significant moments (e.g., popular events)

CODE LOGIC:

1. Data Presentation

- Displays processed highlights in a readable format.

2. Timeline Representation

- Shows highlights over-by-over in sequence.

3. Conditional Tagging

- Adds Popular based on crowd engagement.

4. Structured Output Formatting

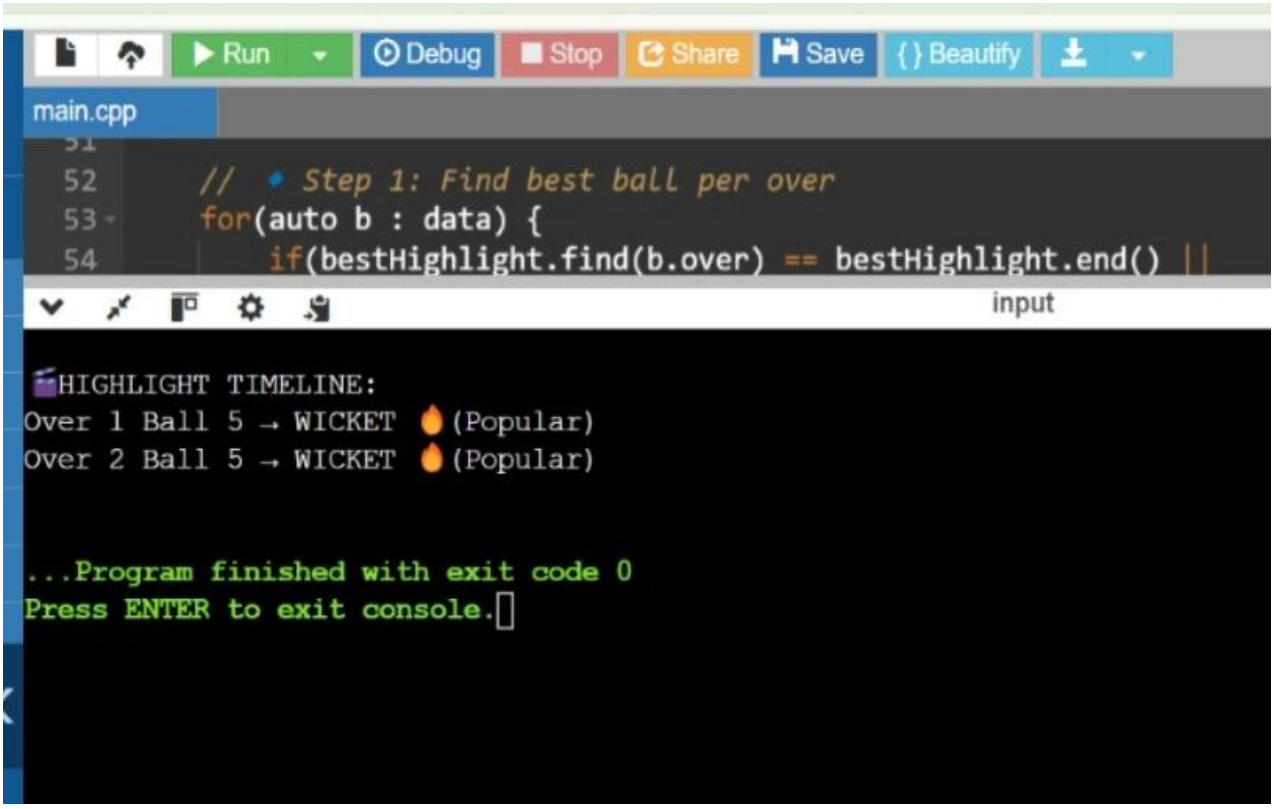
- Uses format:
- Over X Ball Y → EVENT

5. Ordered Display

- Uses map to maintain sorted order of overs.

CODE LINK: [DISPLAY]

SIMULATED OUTPUTS:



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